

“We Aim To **WORK WITH CENTRES OF EXCELLENCE** And Help Them Pragmatically”

Harshad Mehta

Visicon Power



Harshad Mehta and his team at Visicon Power are on the verge of setting up India's first manufacturing plant for Silicon Carbide (SiC) components! The team has already partnered with leading research and academic institutions to ensure that they have access to India's cutting-edge talent, labs, and resources! Here are some interesting excerpts from our interaction with him...

What is the investment that you have come forward with? What are you investing in?

Our plan is to invest in building a total silicon carbide ecosystem in India. This ecosystem will address the weakest links of the silicon carbide industry and all links of the supply chain. For example, starting with the material, device fabrication and followed by device packaging. We are trying to work directly with customers going forward.

Visicon is a subsidiary of a company that I founded in the US. The India plan is completely under the brand name Visicon, and it will address three critical parts of the ecosystem in three phases. The first phase will be silicon carbide epilayering (epi), not only for our own consumption but also for other customers, mostly global. The second phase will be packaging, and the third will be fabrication.

If you look at the process, it is silicon carbide epi first, followed by device fabrication and packaging. But in terms of time, silicon epi and packaging will come earlier than fabrication. It will help us start generating revenue while we are in the process of setting up a semiconductor fab in Baroda.

We acquired a controlling stake in Ruttonsha International in 2006. The technology we are going to be using has been transferred into that name. Silicon Carbide technology will be transferred under the name of Visicon. However, for both Ruttonsha and Visicon Power, Silicon Power will continue to be the umbrella corporation.

Have you planned all the investments in Baroda?

Most of the investments are planned in Baroda. While there will be investments in terms of technology and

market development, 80 to 85 percent have been earmarked for facility development, infrastructure, and training.

Any estimate in terms of units and/or revenue that you expect from the SiC plant in the next five years?

We expect to generate around 300M+ USD in revenue on a cumulative basis over the next 5 years and to make ~50 million devices.

Is this a replica of your business model in the United States?

Yes, but this is not the first time. We have had this kind of a business model setup in India over the last 20 years. In 1994, we acquired a silicon power semiconductor fab from General Electric in the US. In 2006, we transported the technology and the process from there to another fab in Halol, Gujarat. So, we have been following the same business model successfully.

About a year and a half ago, we acquired a fab from Microchip for compound semiconductors. We are trying to follow the same model in a more refined way. Our plan right now is to transfer the technology from our Oregon fab to the one in Baroda.

Can you share details about the key manufacturing equipment you have invested in (type of equipment, brand, model, investment) and any that you plan to invest in—in the next one year?

SiC epitaxial equipment will be from LPE, both pre and post Epi equipment. On the packaging side, we plan to invest in dicing saws, die attach, wire bonder, moulding and tester capital equipment. Currently, our main vendors are from Disco, K&A ;S, ASM, Dage, and a few others. The total investment cost will be over five million dollars.

Have you planned any timelines for the execution of these plans?

Yes, our timelines are very clear. We have applied for the Production Linked Incentive (PLI) scheme in India, which has been approved. We are in the process of applying for Scheme for Promotion of Manufacturing of Electronic Components and Semiconductors (SPECS). As we believe in efficiency a lot, we have already placed an order for epi reactors. Normally, it is a six months lead-time. We had placed an order in September 2020. The reactors are expected to arrive here in March or April 2021.

The equipment will be ready but as a part of the whole process, some of the technical support and services have to come from Italy. There have been some issues in Visa applications due to Covid-19. We are trying to move forward as rapidly as we can.

There has been a lot of talk of why India is not fit for a semiconductor fab, but your take on the same is very different. What are the main reasons?

About India being ready for a semiconductor fab, my answer would be that semiconductors is a very broad space. They are used in consumer electronics to communicate with microcontrollers. Being a part of this space, what distinguishes us from others is the fact that we are small in size. Secondly, our business strategy is very simple—we target only high-end niche products.

We are not looking at high-volume or smaller-node sizes like 7 nm, 24 nm in terms of the technology. My philosophy has never been to follow me-too products. Big names already exist there and we do not want to be there in that market. Our focus is a highly niche, high-performance and high-reliability semiconductor market. The same